Mannan Raj Singh

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Education

Georgia Institute of Technology — Atlanta, GA

B.S. in Computer Science — *Aug 2024 – Dec 2026*

M.S. in Computer Science — *Jan 2027 – Dec 2027*

- **GPA: 3.95/4.00** | Zell Miller Scholar | Concentrations: Intelligence & People | Co-author, PUMP Journal
- Coursework: Data Structures & Algorithms, Objects & Design, Computer Organization, Automata & Complexity, Machine Learning, Artificial Intelligence, Deep Learning, Game AI, Object Oriented Programming

Experience

Amazon — Austin, TX

Software Development Engineer Intern — *Jun 2026 – Aug 2026*

- Led the end-to-end migration of the Save-and-Retrieve workflow for Amazon's Small Business Credit Card application, processing **2,000+ daily requests** for a product serving **1.6M+ customers**; independently owned design, infrastructure, implementation, testing, and production rollout.
- Built fault-tolerant workflows with AWS CDK, Lambda, SNS, SQS, dead-letter queues, retries, and idempotency keys, enabling dual writes to Zima and DIS while preventing duplicate customer reminders.
- Gradually migrated retrieval traffic through two controlled WebLabs, validating **10K+ records with 100% consistency** and **zero observed retrieval failures** across nearly one month of production traffic.
- Integrated directly with three partner teams and built a PII-free DynamoDB datapath view powering downstream BI reporting; configured five CloudWatch alarms with automated SEV-3 ticketing for datastore mismatches.

Mercor — San Francisco, CA

Machine Learning Engineer Intern — *Nov 2025 – Jan 2026*

- Debugged and validated **500+ code-generation tasks** for Meta's Muse AI model, identifying and resolving correctness, runtime, and test-suite failures.
- Built Python and Git-based quality-control automation to detect and flag failed tasks through automated workflows.
- Conducted code reviews for **250+ pull requests**, evaluating code quality, correctness, and task requirements.

Georgia Tech Research Institute — Atlanta, GA

Software Engineer Intern — *May 2024 – Aug 2024*

- Architected a serverless AWS weather mesonet ingesting data from 100+ IoT sensors and handling **5,000+ requests/min** using Lambda, DynamoDB, and event-driven pipelines.
- Built a React and AWS Amplify dashboard that centralized real-time and historical sensor data, enabling researchers to analyze weather patterns across locations and time periods.

Projects

NutriGO | Gamified Nutrition Platform — React/TypeScript, Expo, Node.js, Claude AI

- Built a cross-platform nutrition app using Claude Vision to transform meal photos into ranked collectible cards, allowing users to compare meals and compete in nutrition-based battles.
- Engineered an iMessage webhook for meal logging and a live macro dashboard, earning **3rd place among 50+ teams** at the Claude Builder Club Hackathon (Anthropic × NBC News).

Director's Cut | AI Movie Rewriting Platform — FastAPI, React, pgvector, Neo4j

- Embedded **10,000+ audience reviews across 25+ movies** using Sentence Transformers and pgvector, semantically clustering recurring complaints to identify each movie's most common issues.
- Developed **25+ REST endpoints** and a Neo4j knowledge graph connecting movies, characters, reviews, and themes; used Gemini to generate alternate endings grounded in audience feedback and ElevenLabs for voice narration.

Technical Skills

Languages: Python, Java, JavaScript, TypeScript, C/C++, SQL, HTML/CSS, Bash
Frameworks: React, React Native/Expo, Node.js, Express, FastAPI, Flask, Django, Android SDK
Cloud/Infrastructure: AWS (CDK, Lambda, SNS, SQS, DynamoDB, CloudWatch, Amplify), Docker, Git, Linux, CI/CD
AI/ML: PyTorch, TensorFlow, scikit-learn, pandas, NumPy, Sentence Transformers, LLM APIs (Claude, Gemini)
Databases: DynamoDB, PostgreSQL, Supabase, pgvector, Neo4j

Interests

Golf | Tennis | Basketball | Cooking | Travel | Volunteering | Poker | Logic Puzzles | Music | Photography | Chess